

Lab 09.1: Capgemini Seminar 1

21/11/2025

Abstract

This report summarizes functional safety standards **IEC 61508** (industrial) and **ISO 26262** (automotive), emphasizing risk reduction to a "tolerable level" rather than absolute zero risk.

- **Methodology:** it details the Hazard Analysis and Risk Assessment (HARA) process, which uses Severity, Exposure, and Controllability to assign ASIL rankings (A–D)
- **Framework:** the report outlines the V-Cycle model used for the design, development, and validation of electrical/electronic (E/E) systems
- **Application:** a case study on active suspension and battery management systems demonstrates practical ASIL determination, identifying safety goals ranging from ASIL A to ASIL C

Contents

1.	Introduction.....	4
1.1	IEC 61508 Standard.....	4
1.1.1	Context	4
1.1.2	Functional Safety Definition	4
1.1.3	Passive vs. Active Safety	4
1.1.4	Definition of IEC 61508 Standard	4
1.1.5	Zero Risk Principle.....	5
1.2	Safety Basic Concepts.....	5
1.2.1	Unreasonable/Acceptable risk	5
1.2.2	Terminology	5
1.2.3	Safety Function	5
1.2.4	Fault vs. Failure vs. Hardware Fault Tolerance	5
1.2.5	Safety Integrity Level (SIL)	6
1.2.6	General Safety Development Approach.....	6
1.2.7	Failure Types.....	6
1.2.8	IEC 61508 Workflow	6
1.2.9	Safety Culture	7
1.3	Functional safety related accidents	7
1.3.1	Toyota Accelerator Incident.....	7
2.	ISO 26262 Standard	8
2.1.1	Context	8
2.1.2	Definition of ISO 26262 Standard.....	8
2.1.3	Additions/Exclusions	8
2.1.4	Structure of the ISO 26262	8
2.1.5	Automotive Safety Integrity Level (ASIL).....	9
2.1.6	Basic Concepts - HARA, SG, SM	10
2.1.7	Failure Types.....	10
2.1.8	Points of Failure	10

2.1.9	Acceptable Risk Factors	10
2.1.10	Interpretation of Tables	10
2.1.11	Free decision making.....	11
3.	Guided exercise	12
3.1	Introduction.....	12
3.1.1	Severity Definition and Classification.....	12
3.1.2	Exposure Definition and Classification.....	12
3.1.3	Controllability Definition and Classification	13
3.1.4	ASIL determination table.....	13
3.2	Case study: active suspension system.....	13

1. Introduction

1.1 IEC 61508 Standard

1.1.1 Context

- Emerging Tech: intelligent systems (AI, robotics, IoT) create new complex requirements
- Integration: increasing presence of machines in factories, logistics, mobility, and healthcare requires safety procedures for human interaction
- Safety Definition: freedom from unacceptable risk of physical injury
- IEC: International Electrotechnical Commission; prepares international standards for electrical/electronic technologies

1.1.2 Functional Safety Definition

- **Scope:** part of overall safety related to Equipment Under Control (EUC) and its control system
- Dependency: depends on the correct functioning of the safety-related system
- Achievement: achieved when all specified safety functions satisfy their required safety performance
- Importance: critical in every industry, especially safety-critical ones

1.1.3 Passive vs. Active Safety

- **Passive Safety:**
 - System does not work until called to action
 - Reduces effects of an accident
 - Examples: Airbags, Seatbelts
- **Active Safety:**
 - Works to prevent accidents
 - Activated in response to safety problems or abnormal events
 - Examples: Anti-lock Braking System (ABS), Adaptive Cruise Control (ACC)

1.1.4 Definition of IEC 61508 Standard

- **Scope:** functional safety standard for the lifecycle of electrical, electronic, or programmable electronic (E/E/PE) systems
- Applicability: basic standard applicable to all industries
- Objective: requirements for design, implementation, operation, and maintenance to ensure Safety Integrity Level (SIL)

1.1.5 Zero Risk Principle

- Principle: ensuring safety is key, but recognizing that zero risk is not possible
- Inevitability: if a facility/activity exists, there is a probability of an incident
- Residual Risk: absolute safety does not exist; "Residual risk" always remains
- Relative Safety: products/processes can only be "relatively safe"
- Goal: safety is achieved by reducing risk to a "tolerable level"

1.2 Safety Basic Concepts

1.2.1 Unreasonable/Acceptable risk

- **Residual risk:** risk remaining after safety measures
- **Unreasonable risk:** risk judged unacceptable in a specific context based on societal moral concepts
- **Acceptable risk:** risk where probability and severity are "As Low As Reasonably Practicable" and tolerable
- After risk reduction measures, residual risk should be acceptable
- Acceptability criteria are defined by international commissions/standards

1.2.2 Terminology

- **Hazard:** potential source of harm caused by malfunctioning behaviour
- **Risk:** combination of probability of harm occurrence and severity of that harm
- **Harm:** physical injury, damage to health, property, or environment
- **Tolerable Risk:** target risk level for a specific hazard considering reduction measures

1.2.3 Safety Function

- **Definition:** function implemented by safety-related systems (E/E/PE) to achieve/maintain a safe state for the EUC regarding a specific hazard
- Examples:
 - Positive actions to avoid hazards (e.g., switching off a motor)
 - Preventative functions (e.g., preventing a motor from starting)

1.2.4 Fault vs. Failure vs. Hardware Fault Tolerance

- **Fault:** abnormal condition that can cause an element or item to fail
- **Hardware Fault Tolerance:** hardware capability to resist faults
- **Failure:** termination of the ability of an element to perform a required function

1.2.5 Safety Integrity Level (SIL)

- **Definition:** discrete level (1 to 4) corresponding to safety integrity values
- **Levels:** SIL 4 is highest; SIL 1 is lowest. SIL 0 implies quality development only
- **Targets:** each SIL has a target failure rate (IEC 61508-1)
- **Usage:** used to specify safety integrity requirements for safety functions
- **Clarification:** SIL is a capability limit, not a property of a system/component itself

1.2.6 General Safety Development Approach

- **Steps:**
 - Definition of subject of analysis
 - Hazard Analysis and Risk Assessment (HARA)
 - Safety Function definition and SIL allocation
 - Definition of method/measures according to allocated SIL for each lifecycle part
 - Systematic Failure avoidance and Random HW Failure reduction according to SIL

1.2.7 Failure Types

- **Safety Lifecycle:** activities from concept phase through to decommissioning
- **Systematic Failure:** deterministic failure related to a cause (design, process, documentation error) aiming for elimination
- **Examples:** Calibration, Specification, or Software errors
- **Random Hardware Failure:** unpredictable failure following a probability distribution; not deterministic

1.2.8 IEC 61508 Workflow

- Item definition (function, boundaries, constraints)
- Hazard Analysis and Risk Assessment
- Safety Requirements development (System Level)
- HW/SW safety requirements development
- HW/SW design specification
- Safety Analysis
- Validation & Verification
- User information/safety manual

1.2.9 Safety Culture

- **Definition:** organizational culture placing high importance on safety beliefs, values, and attitudes
- **Bias:** implies a bias to ensure no unreasonable risk
- **Goal:** ensure product safety during operation without failure and in the presence of failure

1.3 Functional safety related accidents

1.3.1 Toyota Accelerator Incident

- Aug 28, 2009. Lexus ES350 accelerated out of control
- Reached 100 mph; 911 call reported "no brakes"
- Outcome: 4 fatalities; crash and fire
- Initial Conclusion: problems deemed mechanical (sticking pedals, pedals under floor mats) or driver error
- Technical Cause: **Unintended acceleration linked to deficiencies in Electronic Throttle Control System (ETCS)**

2. ISO 26262 Standard

2.1.1 Context

- **Functional Safety (FuSa):** addresses systematic and random faults in E/E systems
- Homologation: application not required for homologation but advantageous
- 1st Edition (2011): passenger cars < 3500 kg
- 2nd Edition (2018): extended to all road vehicles (excluding mopeds)
- ISO 26262 Coverage: **Electronic and Electrical systems**

2.1.2 Definition of ISO 26262 Standard

- **Purpose:** defines requirements/guidelines for FuSa in road vehicle E/E systems
- Compliance: mandates documented development for device qualification
- Nature: provisional standard offering methods for safety compliance, allowing technological freedom
- **Objective:** minimize risks in product design/development to prevent hazards
- **Target:** prevent life-threatening failures and achieve acceptable residual risk
- Applies to: Safety-related E/E systems in series production road vehicles (no mopeds)
- Culture Shift: defines shift from Quality Management (ISO/TS 16949) to Safety-Oriented culture
- Requirements: provides validation and confirmation measures to ensure acceptable safety levels

2.1.3 Additions/Exclusions

- Addresses:
 - Alterations to existing systems (tailoring lifecycle)
 - Integration of existing systems
 - Hazards from malfunctioning behaviour of safety-related E/E systems
- **Does Not Address:** hazards like shock, fire, smoke, toxicity, etc., unless caused by E/E malfunction

2.1.4 Structure of the ISO 26262

- **V-Cycle:** illustrates the V-model workflow mapping standard parts to development phases
 - Left side = design/Development; Right side = Validation & Verification
- Total Parts: 12 parts
- Normative: 10 parts (1-9 and 12)

- Guidelines: 2 parts (10 and 11)
- **Part 1 (Vocabulary):** terms, definitions, abbreviations
- **Part 2 (Management):** functional safety management methodology, project-specific info, lifecycle management
- **Part 3 (Concept Phase):** early product development, project definition, safety requirements establishment, initiating lifecycle
- **Part 4 (System Level):** tech safety concepts, system architecture, item integration, testing
- **Part 5 (Hardware Level):** hardware design, architectural metrics, random failure evaluation
- **Part 6 (Software Level):** software safety specs, architecture, unit design/verification, embedded software testing
- **Part 7 (Production/Ops):** production, operation, service, decommissioning requirements
- **Part 8 (Supporting Processes):** Tool qualification, documentation, verification, configuration management
- **Part 9 (Analyses):** ASIL tailoring decomposition, dependent failure analysis, safety analyses
- **Part 10 (Guidelines):** overview and explanations to improve understanding
- **Part 11 (Semiconductors):** guidelines for semiconductor manufacturers and IP suppliers
- **Part 12 (Motorcycles):** adaptation of the standard for motorcycles

2.1.5 Automotive Safety Integrity Level (ASIL)

- **Definition:** risk classification system defined by ISO 26262
- **Purpose:** avoid unreasonable residual risk
- Examples:
 - Airbags/Brakes/Steering = ASIL D (Highest rigor)
 - Rear lights = ASIL A (Lower rigor)
- Levels: A, B, C, D
- Scale: ASIL A (lowest hazard) to ASIL D (highest hazard)
- Visual Map:
 - **ASIL D:** Antilock Braking, Electric Power Steering
 - **ASIL C:** Active Suspension
 - **ASIL B:** Head Lights, Instrument Cluster, Vision ADAS
 - **ASIL A:** Rear Lights
 - **QM:** Quality Management
- ASIL A-D implies unacceptable risk requiring specific safety efforts
- ASIL D: Strictest requirements due to highest risk of injury

- QM: Quality Management development level (standard quality processes sufficient)

2.1.6 Basic Concepts - HARA, SG, SM

- **Hazard Analysis and Risk Assessment (HARA):** method to identify hazards and specify Safety Goals/ASILs to avoid unreasonable risk
- **Safety Goal (SG):** top-level safety requirement resulting from HARA
- **Safety Mechanism (SM):** technical solution (E/E or other) to detect faults or control failures to maintain safe state

2.1.7 Failure Types

- **Systematic Failure:** deterministic, related to cause (design/process), eliminated by change
 - Examples: software error, Specification error
- **Random Hardware Failure:** unpredictable, probabilistic, "just happen"
- **Dependent Failures:** simultaneous/successive occurrences not mathematically independent (Cascading or Common Cause)

2.1.8 Points of Failure

- **Single Point of Failure:** directly violates Safety Goal
- **Double Point of Failure:** violates Safety Goal only in combination with other failures
- **Multiple Point of Failure:** violates Safety Goal only if combined with more than one other failure

2.1.9 Acceptable Risk Factors

- **Probability of Exposure:** frequency of operational situation where accident could occur
- **Severity:** measure of potential injuries (bruises to mortality)
- **Controllability:** driver's capability to take action to prevent injury
- **Judgment:** risk might be unacceptable if difficult to control, even if rare

2.1.10 Interpretation of Tables

- Entry Types:
 - Alternative: marked by number + letter (e.g., 2a, 2b). Choose appropriate combination
 - Consecutive: marked by sequence number (e.g., 1, 2). All methods recommended
- Table Types: Normative or Informative

- The degree of recommendation is categorized as follows
 - "++": highly recommended for the ASIL
 - "+": recommended for the ASIL
 - "o": no recommendation for or against

Methods		ASIL			
		A	B	C	D
1	Functional testing ^a	++	++	++	++
2	Fault injection testing ^b	+	+	++	++
3	Electrical testing ^c	++	++	++	++

- The methods listed in this table are a consecutive entries
- The first and third method « functional testing » and « Electrical testing » are highly recommended for ASIL A, B, C & D
- The second method « Fault injection testing » is recommended for ASIL A & ASIL B, and highly recommended for ASIL C & ASIL D

2.1.11 Free decision making

- Concept: **standard allows deciding not to perform an activity or use a method**
- Constraint: must provide robust rationale
- Rationale Examples: activity done by others (OEM/Supplier), covered by other analysis, or method not required for the specific ASIL
- Organizations must have an operational Quality Management System (QMS)

3. Guided exercise

3.1 Introduction

- **Topic:** active Suspension System HARA (ISO 26262 Part 3 Annex B)
- Purpose: derive Safety Goal and ASIL
- Process:
 - List malfunctions (interview developer)
 - Evaluate hazardous consequences
 - Select use cases (driving, road, weather, traffic)
 - Evaluate Severity, Exposure, Controllability

3.1.1 Severity Definition and Classification

- **Definition:** estimate of the extent of harm to individuals in a hazardous event
- S0: no injuries
- S1: light/moderate injuries
- S2: severe/life-threatening injuries (survival probable)
- S3: life-threatening/fatal injuries (survival uncertain)

3.1.2 Exposure Definition and Classification

- **Definition:** state of being in an operational situation hazardous if coincident with failure
- E1 to E4: Classified by operational situations (Road surface, vehicle state, maneuvers)
- Exposure probability classes:
 - E1: jump start
 - E2: snow/Ice, Trailer attached
 - E3: wet road, Heavy traffic
 - E4: accelerating, Lane change
- Exposure duration classes:
 - E1: very low probability (Not specified)
 - E2: low probability (<1% operating time)
 - E3: medium probability (1% to 10% operating time)
 - E4: high probability (>10% operating time)
- Exposure frequency classes:
 - E1: < once a year
 - E2: few times a year
 - E3: once a month or more

- E4: almost every drive

3.1.3 Controllability Definition and Classification

- **Definition:** Ability to avoid specified harm through timely reaction (driver, passengers, bystanders)
- Controllability classes
 - C0: controllable in general
 - C1: simply controllable (>99% of drivers can avoid harm)
 - C2: normally controllable (>90% of drivers can avoid harm)
 - C3: difficult to control/uncontrollable (<90% of drivers can avoid harm)

3.1.4 ASIL determination table

- Matrix: Combines Severity (S), Exposure (E), and Controllability (C) to output QM, A, B, C, or D
- Example: S3 + E4 + C3 = ASIL D

3.2 Case study: active suspension system

- Steps
 - Brainstorming on hazards
 - Malfunctioning behaviour list → what could go wrong
 - Hazard optimization summarizing specific malfunctions in classes
 - Assign severity, exposure and controllability classes
- Summary (BCM = battery charge management system)
 - **BCM_SG1:** Avoid Fire/thermal runaway on MVB -> **ASIL C** -> Safe State: No power flow
 - **BCM_SG2:** Avoid Human Electrocution -> **ASIL B** -> Safe State: 48V Bus discharged < 60V
 - **BCM_SG3:** Avoid fire on 48V cables -> **ASIL A** -> Safe State: Devices disconnected/Switched Off